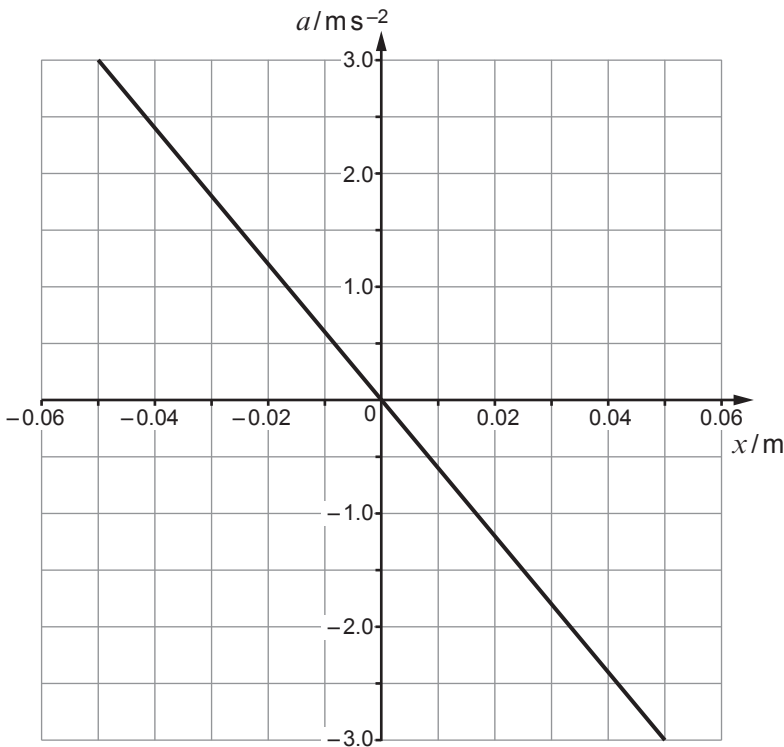


Examiner
only

3. A small object of mass 0.30 kg oscillates at the end of a spring. The graph shows how its acceleration, a , depends on its displacement, x , from a fixed point.



(a) (i) Identify the **two** characteristics **of the graph** that show that the object is oscillating with *simple harmonic motion*. [2]

.....

.....

.....

.....

(ii) Use the graph to show that the angular frequency, ω , of the oscillation is approximately 7.7 rad s^{-1} . [3]

.....

.....

.....

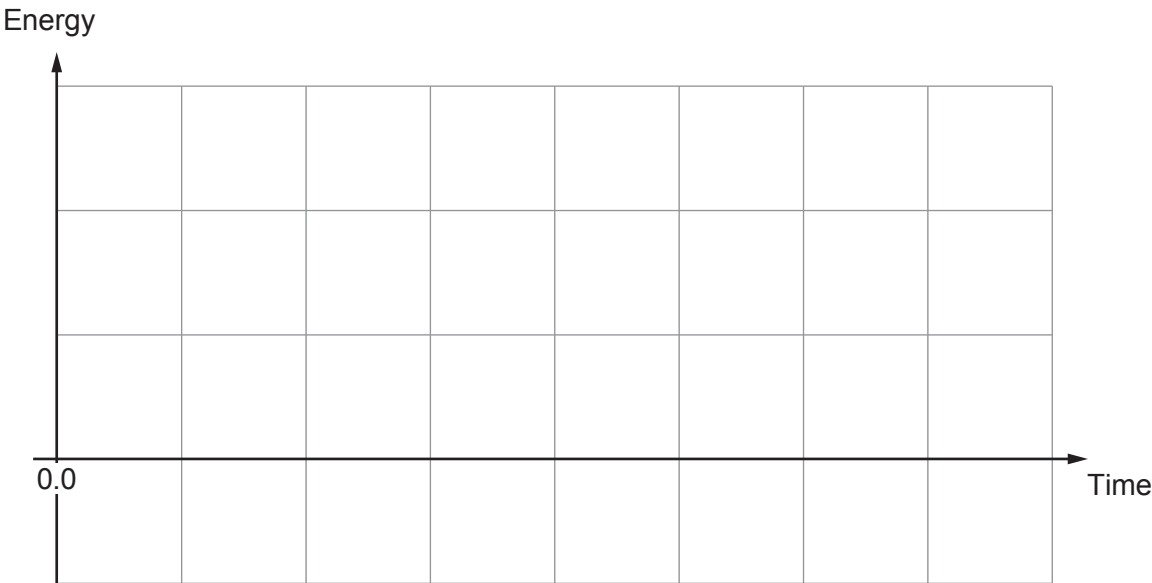
.....

.....



Examiner
only

- (iii) Calculate:
- I. the period of oscillation; [1]
-
-
- II. the spring constant; [2]
-
-
-
- III. the velocity of the object as it passes through the centre of oscillation. [2]
-
-
-
- (iv) The object was released at the displacement of -0.05 m at time $t = 0$. Sketch on the axes below:
- I. the potential energy of the system (label the sketch PE),
- II. the kinetic energy of the system (label the sketch KE).
- Draw both sketches **over one cycle** of the undamped oscillation. (No values are required on the axes.) [3]



(b) Explain what is meant by the terms *damping* and *resonance* **and** discuss their importance in real systems giving an example for each.

[6 QER]

Examiner
only

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

